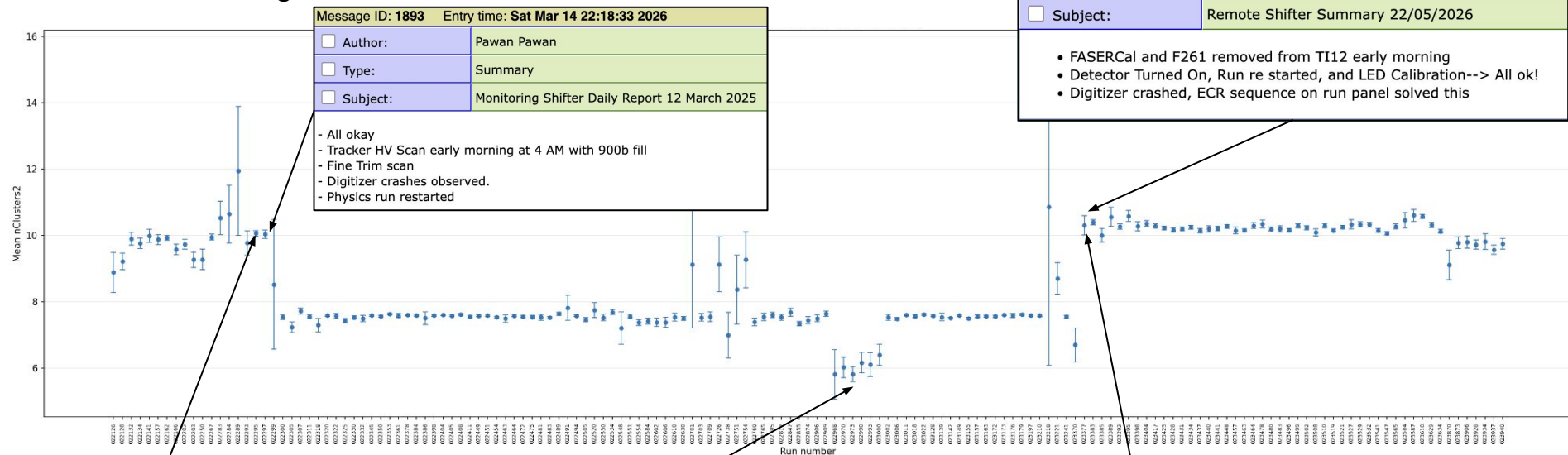


Station 2 Issue

Tracker Meeting
25/08/26

Oscar Martinez

Digitizer Crashed: Run: 22299 on 12/03/26



FTS@800b

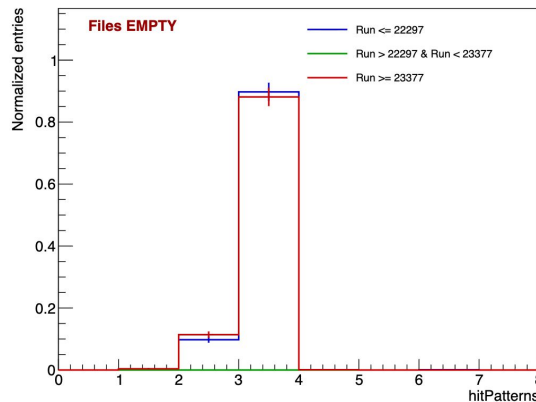
1.2 TeV runs

FASERnu Removal / Start of Ion Run
Run: 023377 on 23/05/26.
Power Cycled detector.

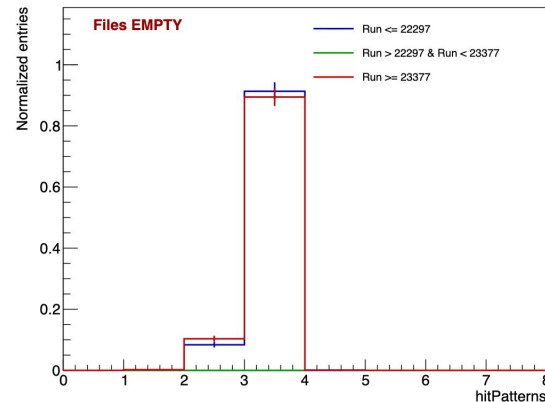
Timing Patterns

- As we ran Fine Time Scan one run just after the crash, we suspected maybe the timing information is wrong.
 - Maybe going into the 1XX pattern?
- Using fine time scan ntuple maker we can easily extract timing information without 01X restrictions.
- Unfortunately, does not seem to be the case, as no timing information is found, i.e no early or late tracks...
- This implies the track information is not on a different timing bin.
- All modules timing distributions look fine.

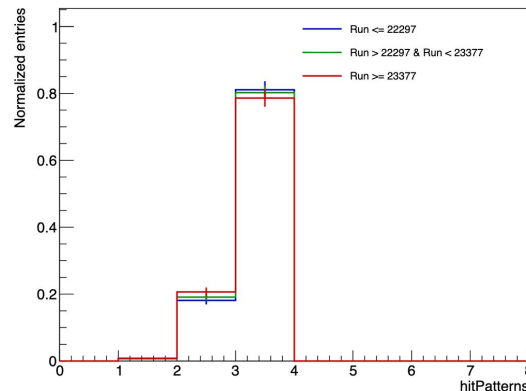
station=2, layer=2, column=1, row=3, side=0



station=2, layer=2, column=1, row=3, side=1



station=2, layer=1, column=1, row=3, side=0



1: Circular segment above chord $y = d$

For a chord at distance d from the center, the segment area is

$$A_{\text{seg}}(d) = r^2 \cos^{-1}\left(\frac{d}{r}\right) - d\sqrt{r^2 - d^2} \quad (1)$$

With $d = 60$, $r = 90$:

$$A_{\text{seg}} = (90)^2 \cos^{-1}\left(\frac{60}{90}\right) - 60\sqrt{90^2 - 60^2} \quad (2)$$

$$= 8100 \times 0.8411 - 60 \times 67.08 \quad (3)$$

$$\approx 6813 - 4025 \quad (4)$$

$$\approx 2788 \text{ mm}^2 \quad (5)$$

2: Restrict to $x \geq 0$ half

The segment is symmetric about $x = 0$; the dead module only covers $x \geq 0$:

$$A_{\text{corner}} = \frac{A_{\text{seg}}}{2} \approx \frac{2788}{2} \approx 1394 \text{ mm}^2 \quad (6)$$

3: Total circle area

$$A_{\text{circle}} = \pi r^2 = \pi(90)^2 \approx 25447 \text{ mm}^2 \quad (7)$$

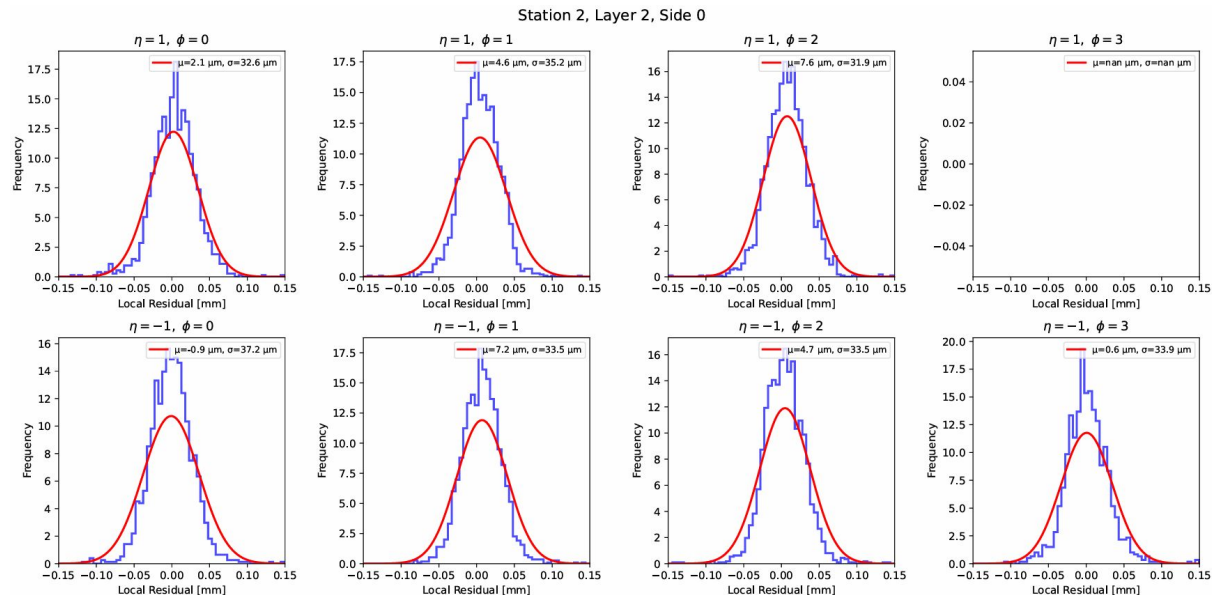
4: Fraction lost

$$f = \frac{A_{\text{corner}}}{A_{\text{circle}}} = \frac{1394}{25447} \approx 0.0548 \Rightarrow \boxed{5.5\%} \quad (8)$$

- 5.5% of the FV covered by the “dead” module.
- Assuming uniform distribution of tracks (we know its not): 5.5% of tracks will have -2 DoF.
- Likely to impact tracking performance (momentum, charge id).
- Effect to be quantified.

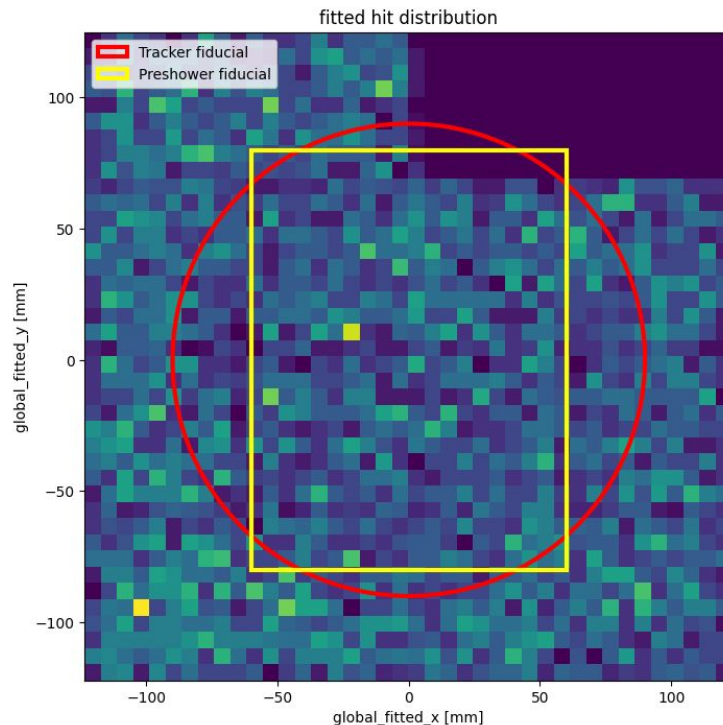
Backup

- While performing unbiased residuals for 2026, Haruhi spotted one module which had empty information.
- After running with more statistics and different runs problem seemed to persist.
- More investigation done after this...

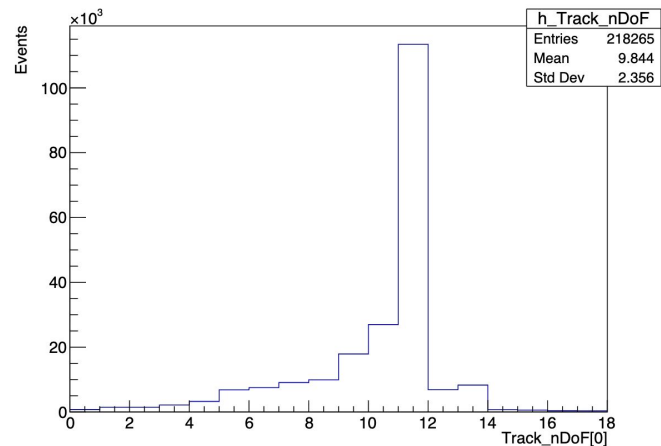
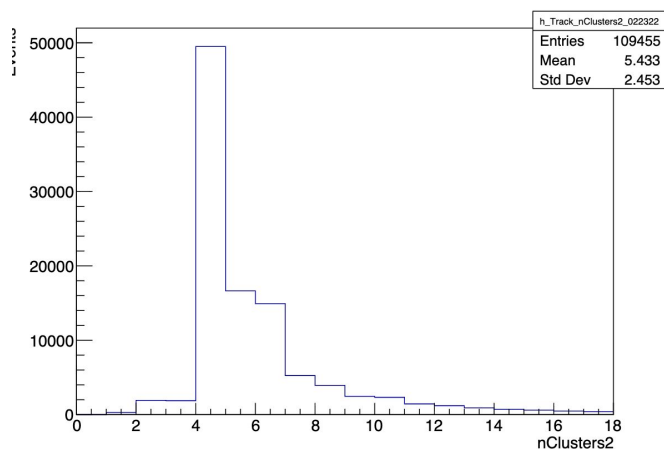


Hitmap

- Using alignment ntuples (information from all modules) we can plot hit map of only ST2L2.
 - Something which is not possible with current prompt reco at the moment.
- Module seems to not be recording any information ...
- All other modules on that layer and all stations seem to be working fine.



- We can observe this deficit while looking at the number of clusters and DoF for tracks passing through this module.
- nClusters peaks at 4 (instead of 6) and DoF peaks at 11 (expected 13).



```
Track_module_eta0[0] == 1 &&  
Track_module_phi0[0] == 3 &&  
longTracks > 0
```


- Pawan also spotted no strips activated.
- Since this info is from raw, maybe it's not recoverable?
- Full trim scan and later calibrations suggest this module is now fine after the power cycle post-FASERnu removal.
- Michaela suggested looking at the icc for the LV during this period, however I don't spot anything particularly different during this period on grafana.